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ScienceDirect
Intermetallics 15 (2007) 1401–1408
Intermetallics
www.elsevier.com/locate/intermet

Reassessment of the Ce–Ni binary system supported by key experiments and *ab initio* calculations

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Received 16 February 2007; received in revised form 4 April 2007; accepted 8 April 2007
Available online 7 June 2007

Abstract

The Ce–Ni system is reassessed by means of CALPHAD (CALculation of PHAse Diagram) approach, supplemented with decisive experiments and enthalpies of formation computed via VASP (Vienna *ab initio* simulation package) code. Fourteen alloys are prepared in order to check the general feature of the established Ce–Ni phase diagram and to provide new phase diagram data for the refinement of previous modeling. The phase identification for both as-cast and annealed states is performed by means of X-ray diffraction technique, and the phase transition temperatures are measured by differential thermal analysis. Electron probe microanalysis is employed to determine two tie lines between $CeNi_5$ and (Ni). The enthalpies of formation for $CeNi_2$ and $CeNi_5$ are calculated using VASP code with a desire to clarify the discrepancies among the literature data. A set of self-consistent thermodynamic parameters for the Ce–Ni system are finally obtained. Significant improvements have been made, compared with the previous assessments.
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Keywords: A. Rare-earth intermetallics; B. Thermal properties; E. Ab initio calculations; E. Phase diagram, prediction

1. Introduction

The Ce–Ni system is widely used as hydrogen storage material in engines and automobiles due to its high hydrogen absorption ability. In addition, this system is one of the important sub-binary systems in multi-component Al-based and Mg-based alloys in which amorphous alloys with high strength and good ductility can be synthesized [1,2]. Consequently, many researchers have paid extensive attention to this important system for the development of hydrogen cells and bulk-glass materials. Du et al. [3], Palumbo et al. [4] and Pisch et al. [5] have carried out the thermodynamic modeling of the Ce–Ni system. In spite of this, a thermodynamic reassessment for the Ce–Ni system is still needed mainly due to the following three reasons. Firstly, the measured homogeneous range of the $CeNi_5$ compound [6] was not considered in the two previous assessments [3,4]. Secondly, both Du et al. [3] and Palumbo et al. [4] ignored the experimental heat content for $CeNi_5$ [7]. Thirdly, although Pisch et al. [5] took into account the above experimental information and a good description of the Ce–Ni system is obtained, the complex quasi-chemical model cannot be combined with the thermodynamic description of the published binaries to calculate the multi-component system. For example, the Al–Ce [8] and Al–Ni [9] systems with substitutional model for liquid have been modeled. Thus, the quasi-chemical model used for liquid phase in the Ce–Ni system cannot be combined with the description of the liquid phases in the Al–Ce and Al–Ni systems to calculate the phase equilibria and thermodynamic properties in the Al–Ce–Ni ternary system.

The present work is devoted to obtain an optimized data set for the Ce–Ni system via a combination approach of CALPHAD, key experiments and *ab initio* calculations for enthalpy of formation.

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doi:10.1016/j.intermet.2007.04.004

2. Evaluation of experimental data

The Ce–Ni system contains six intermetallic compounds (Ce_7Ni_3, CeNi, $CeNi_2$, $CeNi_3$, Ce_2Ni_7, and $CeNi_5$). This system has been previously evaluated by Massalski et al. [10], Nash and Tung [11]. In this section, all of the experimental phase diagram and thermodynamic data available for the Ce–Ni binary system are briefly evaluated. Table 1 presents the phase diagram and thermodynamic data, which are considered in the present thermodynamic modeling.

Table 1
Summary of the experimental data included in the present optimization

Type of data	Experimental technique[a]	Reference
(δCe) liquidus	DTA and MA	[12]
	XRD, DTA and MA	[16]
(γCe) liquidus	DTA and MA	[12]
	XRD, DTA and MA	[16][b]
	XRD and DTA	This work[b]
Ce_7Ni_3 liquidus	DTA and MA	[12]
	XRD, DTA and MA	[16][b]
CeNi liquidus	DTA and MA	[12]
	XRD, DTA and MA	[16]
	Density measurement	[20]
	XRD and DTA	This work[b]
$CeNi_2$ liquidus	DTA and MA	[12]
	XRD, DTA and MA	[16][b]
	XRD and DTA	This work[b]
$CeNi_3$ liquidus	DTA and MA	[12]
Ce_2Ni_7 liquidus	Thermal analysis	[21]
$CeNi_5$ liquidus	Reactive diffusion method	[6][b]
	XRD, DTA and MA	[16]
	DTA and MA	[5]
	Thermal analysis	[21][b]
	XRD and DTA	This work[b]
$CeNi_5$ solvus	Reactive diffusion method	[6]
	SEM/EDX	[5]
	XRD, DTA and EPMA	This work
(Ni) liquidus	Reactive diffusion method	[6]
	DTA and MA	[12]
	XRD, DTA and MA	[16]
	XRD and DTA	This work
Eutectoid temperatures	XRD, DTA and MA	[16]
Eutectic temperatures	XRD and DTA	This work
Peritectic temperatures	XRD and DTA	This work
Congruent temperatures	Reactive diffusion method	[6]
	XRD, DTA and MA	[16]
	XRD and DTA	This work
Enthalpies of formation	Aneroid isoperibol calorimetry	[4]
	High-temperature calorimetry	[4]
	Tin solution calorimetry	[7]
	Dissolution calorimetry	[23]
	High-temperature solution calorimetry	[24]
	Ab initio calculation	This work[b]
	Drop solution calorimetry	[5]
Enthalpies of melting	Aneroid isoperibol calorimetry	[4]
	High-temperature calorimetry	[4]
Heat content	Tin solution calorimetry	[7]

[a] DTA = differential thermal analysis; MA = metallographic analysis; XRD = X-ray diffraction; SEM = scanning electron microanalysis; EDX = energy dispersive X-ray analysis; EPMA = electron probe microanalysis.
[b] A relative high weight was attached to those data in the present thermodynamic optimization.

2.1. Phase diagram data

Many authors contributed to the identification of the compounds in the Ce–Ni system. By means of thermal analysis and metallography, the Ce–Ni phase diagram was first established by Vogel et al. [12], and six intermetallic compounds are accordingly reported: Ce_3Ni, CeNi, $CeNi_2$, $CeNi_3$, $CeNi_4$, and $CeNi_5$. Subsequent investigators [13–17] confirmed the stoichiometries of CeNi, $CeNi_2$, $CeNi_3$ and $CeNi_5$, but modified the stoichiometries of Ce_3Ni and $CeNi_4$ phases to be Ce_7Ni_3 and Ce_2Ni_7, respectively. These latter recommended stoichiometries are accepted in the present modeling.

Due to the facts that the terminal phases, (γCe), (δCe) and (Ni), show very limited solubility ranges and pure Ce is prone to be oxidized, it is difficult to precisely determine the phase boundaries associated with the terminal phases. The solubility of Ce in (Ni) at 400 and 1200 °C was approximately determined by Duisemaliev [18] using metallography, microhardness and electrical resistance measurements. The solubilities of Ni in (γCe) and (δCe) were estimated in a review reported by Gschneidner and Verkade [19]. These approximate data [18,19] are not utilized in the present optimization.

The phase relationships in the whole composition range were reinvestigated by Gebhart et al. [16], who employed X-ray diffraction (XRD), differential thermal analysis (DTA) and metallography methods. This type of data is considered to be reliable and, therefore, adopted in the present modeling.

Using the density measurements, Perkins et al. [20] measured the phase transformation temperatures at 34.8, 36.6, and 40.2 at.% Ni. Those data are not adopted in the current assessment since the impurities in the alloys not only yield a lower melting temperature of 787 °C for pure Ce, compared with the generally accepted temperature of 799 °C, but also result in a large difference between nominal and practical compositions for the prepared alloys.

The liquidus from 58 at.% Ni to pure Ni reported by Qi et al. [21] is mainly based on the thermal analysis. Using the reactive diffusion method, Kim et al. [6] determined the homogeneous range of the $CeNi_5$ with the widest homogeneity of 82.4–86.5 at.% Ni at 1573 K. Since these data [6,21] are the only detailed experimental information on the phase relations at high temperatures, they are included in the optimization.

In the work of Pisch et al. [5], the phase equilibria in the Ni-rich part of the Ce–Ni system were determined by means of DTA, metallography and SEM/EDX. The experimental results are used to verify the final thermodynamic modeling.

2.2. Thermodynamic data

Several groups of investigators have contributed to the determination of the enthalpy of formation of $CeNi_5$: Colinet and Pasturel [22] and Meyer-Liautaud et al. [23] using dissolution calorimeter; Yamaguchi et al. [7] employing tin solution calorimeter; Reddy et al. [24] utilizing high-temperature solution calorimeter; Guo and Kleppa [25,26] through high-temperature direct synthesis calorimeter; Wang et al. [27] and Pisch et al. [5] employing Al drop solution calorimeter. It is worth mentioning

that Wang et al. [27] and Pisch et al. [5], who are from the same group, reported the same experimental value for the enthalpy of formation of $CeNi_5$. Since the present *ab initio* calculated value for the enthalpy of formation of $CeNi_5$ could clarify the discrepancies among the reported experimental results [5,7,22–24,27], they are adopted in the optimization with a relatively high weight.

Yamaguchi et al. [7] reported the heat content of $CeNi_{5.13}$ over the temperature range of 800–1400 °C by means of a drop calorimeter. Reddy et al. [24] determined the enthalpies of formation for $CeNi_2$ at 298.15 K using a high-temperature solution calorimeter. Guo and Kleppa [25,26] determined the enthalpies of formation for CeNi at 298.15 K using a high-temperature direct synthesis calorimeter. These data [7,24–26] have been used in the present modeling due to the high purities of starting materials, the well-described experimental procedures and the consistency among the different experimental methods.

By means of a high-temperature calorimeter, the enthalpy of mixing in the liquid phase has been determined at 1487 °C by Nikolaenko and Vlasova [28] in the whole composition range and at 1597 °C by Sudavtsova et al. [29] in the Ni-rich side up to 20 at.% Ce. The low temperature heat capacity and the entropy at 298.15 K have been reported for $CeNi_2$ and $CeNi_5$ [30,31]. These two types of data are not used in the optimization but instead have been considered for a posterior validation of the assessed data.

Enthalpies of melting for three eutectic alloys in the Ce-rich side of the Ce–Ni system have been determined by Palumbo et al. [4] using calorimetric measurements. These values are considered to be reliable and thus utilized in the present optimization.

3. Experimental procedure

For the purpose of checking the general feature of the established Ce–Ni phase diagram, 14 alloys were arc-melted from elements with purities of 99.95 mass% Ce and 99.99 mass% Ni (both from Beijing Mountain Technical Development Center for Non-ferrous Metals, P.R. China). The as-cast alloys were encapsulated in evacuated silica capsules with a residual argon pressure of 10^{-3} bar, annealed at seven temperatures (440, 580, 690, 880, 940, 1050 and 1100 °C) for 3 weeks, followed by water-quenching. The compositions of the alloys are listed in Table 2. Since the weight losses during arc melting were generally less than 1 mass%, no chemical analysis for the alloys was conducted.

It is necessary to emphasize that, in order to minimize the extent of oxidation before arc melting, solid pieces of Ce and Ni are vacuum-packed in Al-foils (thickness of 0.2 mm) with a pressure of less than 10^{-2} bar by employing the vacuum-packed machine (DZ-350/2P, Shanghai Qingpa Package Machine Company, P.R. China). In addition, during the alloy preparation, special attentions were paid to ensure that there is no interval among the processes of weighting, package, and melting. Such a way to avoid the possible oxidation was not taken by previous investigators [4–8,12,16–31].

Table 2
Summary of the phases and phase transition temperatures for the samples in the Ce–Ni system annealed at the specified temperatures for 3 weeks

No.	at.% Ni	Annealing temperature (°C)	Phase[a]	Phase transition temperature[b] (°C)
1	93	1100	$CeNi_5$, (Ni)	1214, 1268
2	88	1050	$CeNi_5$, (Ni)	1214, 1267
3	85	940	$CeNi_5$	1206, 1290
4	82	940	Ce_2Ni_7, $CeNi_5$	1197, 1328
5	80	940	Ce_2Ni_7, $CeNi_5$	1333
6	78	880	Ce_2Ni_7, $CeNi_5$	1064, 1305
7	75	690	$CeNi_3$	957, 1079
8	72	690	$CeNi_2$, $CeNi_3$	813, 956, 1076, 1119
9	67	580	$CeNi_2$	813, 953
10	60	580	CeNi, $CeNi_2$	658, 804
11	50	440	CeNi	659, 679
12	40	440	Ce_7Ni_3, CeNi	487, 589
13	30	440	Ce_7Ni_3	482
14	20	440	Ce_7Ni_3, (γCe)	481, 495

[a] Phases were identified with XRD technique.
[b] Obtained from DTA measurement with a heating rate of 5 °C/min.

Phase identification was carried out by means of XRD (Rigaku D/max2550VB, Japan) with Cu Kα1 radiation and employing high purity Si as an internal standard.

DTA apparatus (DSC404C, NETZSCH, Germany) was used to measure the phase transition temperatures. The sample holder material was pure Al_2O_3, and solid samples with weights of about 50–100 mg were taken from the annealed samples. The measurements were performed between room temperature and 1450 °C with a heating and cooling rates of 5 °C/min in an argon atmosphere. The temperature was measured with Pt–Pt/Rh thermocouples and calibrated to the melting temperatures of Al (660.32 °C), Ag (961.78 °C), Au (1064.18 °C) and Ni (1455.15 °C). In the temperature range examined, the accuracy of the temperature measurement was estimated to be ±2 °C. The invariant reaction temperature was determined from the onset of the first thermal effect during the heating step, and the peak maximum of the last thermal effect on heating was taken for the liquidus.

In order to confirm the homogeneity range of the $CeNi_5$ compound reported by Kim et al. [6], two alloys (Ce_7Ni_{93} and Ce_2Ni_{98}, in at.%), which were annealed at 1100 and 1050 °C for 3 weeks, were grinded, polished, and then subjected to electron probe microanalysis (EPMA, JXA-8800R, JEOL, Japan).

4. *Ab initio* calculations for enthalpy of formation

In order to clarify the discrepancies among the literature data concerning the enthalpies of formation for $CeNi_2$ and $CeNi_5$, Vienna *ab initio* simulation package (VASP) [32–34] is employed to calculate the enthalpies of formation for these two compounds.

The calculations were performed in the framework of density functional theory (DFT) using the generalized gradient approximation (GGA) of Perdew and Wang [35,36]. The ground-state energies of the compound and its constituents were calculated

by using VASP [32–34] with projector augmented wave (PAW) pseudopotentials [32]. Brillouin-zone integrations were performed on a certain grid of Monkhorst–Pack special points [37], i.e. $11 \times 11 \times 11$ for $CeNi_5$ and $15 \times 15 \times 15$ for $CeNi_2$. Full relaxation was taken into account for $CeNi_5$.

The energy of formation for a compound A_xB_y is obtained by the following equation:

$$\Delta E^{\text{form}} = E^{A_xB_y} - xE^{A} - yE^{B} \tag{1}$$

where $E^{A_xB_y}$, E^{A}, and E^{B} are the total energies of A_xB_y, A, and B, respectively. In the present work, A_xB_y represents the compound $CeNi_2$ (*cF*24, *Fd*3*m*) or $CeNi_5$ (*hP*6, *P*6/*mmm*) at its stoichiometric composition. A and B refer to (Ni) and (γCe), respectively.

5. Thermodynamic models

5.1. Unary phases

The Gibbs energy function ${}^0G_i^{\varphi}(T) = G_i^{\varphi}(T) - H_i^{\text{SER}}$ for element i (i = Ce or Ni) in the phase φ (φ = (Ni), (γCe), (δCe) or liquid) is described by an equation of the form

$$ {}^0G_i^{\varphi}(T) = a + bT + cT\ln(T) + dT^2 + eT^{-1} + fT^3 + gT^7 + hT^{-9} \tag{2}$$

in which H_i^{SER} is the molar enthalpy of the element i at 298.15 K and 1 bar in its standard element reference (SER) state, and T is the absolute temperature. The last two terms in Eq. (2) are used only outside the ranges of stability [38], the term gT^7 for a liquid below the melting point and hT^{-9} for solid phases above the melting point. In the present modeling, the Gibbs energies as a function of temperature are taken from the SGTE compilation by Dinsdale [39].

5.2. Solution phases

The φ ((Ni), (γCe), (δCe) or liquid) phase is described with a substitutional solution model:

$$G_m^{\varphi} = x_{Ce}\,{}^0G_{Ce}^{\varphi} + x_{Ni}\,{}^0G_{Ni}^{\varphi} + RT\left(x_{Ce}\ln x_{Ce} + x_{Ni}\ln x_{Ni}\right) + {}^{ex}G_m^{\varphi} + {}^{mag}G_m^{\varphi} \tag{3}$$

where x_i is the mole fraction of element i (i = Ce or Ni) and ${}^0G_i^{\varphi}$ the Gibbs energy of element i with the state of φ. The third term in Eq. (3) corresponds to the contribution to Gibbs energy from the ideal entropy of mixing, the forth term is the excess Gibbs energy, and the last term is the magnetic contribution to the Gibbs energy.

The excess Gibbs energy is described by the Redlich–Kister polynomial [40]:

$$ {}^{ex}G_m^{\varphi} = x_{Ce}x_{Ni}\left[\left({}^{\varphi}a^0_{Ce,Ni} + {}^{\varphi}b^0_{Ce,Ni}T\right) + (x_{Ce} - x_{Ni})\left({}^{\varphi}a^1_{Ce,Ni} + {}^{\varphi}b^1_{Ce,Ni}T\right)\right] \tag{4}$$

The parameters denoted as ${}^{\varphi}L_{ij} = {}^{\varphi}a_{i,j} + {}^{\varphi}b_{i,j}T$ (i, j = Ce or Ni) are the interaction parameters for the binary system.

For liquid, ${}^{mag}G_m^{\varphi}$ is equal to zero. For (Ni), (γCe) or (δCe), there is a magnetic contribution to Gibbs energy. This magnetic contribution is given by the following equation:

$$ {}^{mag}G_m^{\varphi} = RT\ln(\beta^{\varphi} + 1)g(\tau^{\varphi}) \tag{5}$$

where τ is T/T^*, T^* being the critical temperature for magnetic ordering (Curie temperature T_C for ferromagnetic materials or Neel temperature T_N for antiferromagnetic materials), and β is the average magnetic moment per atom expressed in Bohr magnetrons. The polynomial $g(t)$ has been defined by Hillert and Jarl [41].

Due to the negligible solubilities between Ce and Ni, none of the magnetic interaction parameters between Ce and Ni have been used in the present modeling.

5.3. Intermetallic compounds

The intermetallic compounds, Ce_7Ni_3, CeNi, $CeNi_3$ and Ce_2Ni_7, have been described as stoichiometric phases, and the Gibbs energy per mole of $Ce_{(1-x)}Ni_x$ compound is given as follows:

$$ {}^0G_m^{Ce_{(1-x)}Ni_x} - (1-x)H_{Ce}^{\text{SER}} - xH_{Ni}^{\text{SER}} = A + BT + (1-x)\,{}^0G_{Ce}^{fcc} + x\,{}^0G_{Ni}^{fcc} \tag{6}$$

in which A and B are the parameters to be evaluated in the course of optimization. The parameter ${}^0G_i^{fcc}$ is the Gibbs energy of pure species i (i = Ce or Ni) in its stable *fcc* structure.

Following a suggestion by Ansara et al. [42], a two-sublattice model, $(Ce, Ni)_1(Ce, Ni)_2$, has been adopted for the description of Laves phase $CeNi_2$. The Gibbs energy of $CeNi_2$ per mole of formula unit can be expressed as:

$$\begin{aligned} {}^0G_m^{CeNi_2} - H_{Ce}^{\text{SER}} - 2H_{Ni}^{\text{SER}} = {} & y'_{Ce}y''_{Ce}\,{}^0G_{Ce:Ce}^{CeNi_2} + y'_{Ni}y''_{Ni}\,{}^0G_{Ni:Ni}^{CeNi_2} \\ & + y'_{Ce}y''_{Ni}\,{}^0G_{Ce:Ni}^{CeNi_2} + y'_{Ni}y''_{Ce}\,{}^0G_{Ni:Ce}^{CeNi_2} \\ & + RT\left(y'_{Ce}\ln y'_{Ce} + y'_{Ni}\ln y'_{Ni}\right) \\ & + 2RT\left(y''_{Ce}\ln y''_{Ce} + y''_{Ni}\ln y''_{Ni}\right) \\ & + y'_{Ce}y'_{Ni}y''_{Ce}L^0_{Ce,Ni:Ce} \\ & + y'_{Ce}y'_{Ni}y''_{Ni}L^0_{Ce,Ni:Ni} \\ & + y'_{Ce}y''_{Ce}y''_{Ni}L^0_{Ce:Ce,Ni} \\ & + y'_{Ni}y''_{Ce}y''_{Ni}L^0_{Ni:Ce,Ni} + \ldots \end{aligned} \tag{7}$$

in which y'_{Ce} and y'_{Ni} are the site fractions of Ce and Ni on the first sublattice, and y''_{Ce} and y''_{Ni} on the second sublattice. The four parameters denoted as ${}^0G_{*:*}^{CeNi_2}$ (* means the species, Ce or Ni) are expressed relative to the Gibbs energies of (Ni) and (γCe) at the same temperature. The L^0 parameters represent the interactions primarily within each sublattice.

The $CeNi_5$ phase has been described with a two-sublattice model in view of the homogeneity range reported by Kim et al. [6]. The model used is (**Ce**, Ni)$_1$(Ce, **Ni**)$_5$, where the first sublattice is mainly occupied by Ce and the second by Ni. Since the experimental data on heat content of $CeNi_5$ over the wide temperature ranges are available [7], it is preferable to express the Gibbs energies of the $CeNi_5$ compound relative to the SER state. The Gibbs energy of $CeNi_5$ per mole of formula unit is expressed as:

$$G_{\mathrm{m}}^{\mathrm{CeNi_5}} - H_{\mathrm{Ce}}^{\mathrm{SER}} - 5H_{\mathrm{Ni}}^{\mathrm{SER}} = a + bT + cT\ln T + dT^2 + eT^{-1} \quad (8)$$

The coefficients c–e can be evaluated directly from the heat capability expression.

6. Results and discussion

The phase transition temperatures for the prepared alloys are listed in Table 2. The reproducible measurements confirm the general feature of the established Ce–Ni phase diagram [3,4]. For the temperatures of several invariant reactions, however, noticeable deviations are observed. Most prominently, the present DTA measurement yields the reaction temperature of 813 °C for the invariant reaction, L + $CeNi_3$ ↔ $CeNi_2$, which is lower by 17 °C than the previous ones [12,16,17,19]. The present experimental value is preferable since the extent of oxidation is controlled to be negligible due to the vacuum packing during the preparation of the alloy.

The XRD data confirmed all established Ce–Ni intermetallic phases and their crystal structures. The observed lattice parameters match closely those reported in Ref. [4] and thus are not presented here.

The optimization of the thermodynamic parameters in the Ce–Ni system was performed with the PARROT module of the Thermo-calc software [43]. The step-by-step optimization procedure described by Du et al. [44] was adopted in the present assessment. In the optimization, each piece of experimental information is given a certain weight. The weights were changed systematically during the assessment until most of the experimental data were accounted for within the claimed uncertainty limits.

For the liquid, at least ${}^{L}a^{0}_{\mathrm{Ce,Ni}}$ and ${}^{L}b^{0}_{\mathrm{Ce,Ni}}$ in Eq. (4) can be adjusted because the liquidus was measured accurately over the whole composition range. In the present modeling, it was found that ${}^{L}a^{1}_{\mathrm{Ce,Ni}}$ and ${}^{L}b^{1}_{\mathrm{Ce,Ni}}$ should also be introduced in order to describe the properties of the liquid phase satisfactorily.

Since the solubilities of Ce in (Ni), and Ni in (γCe) and (δCe) reported by Duisemaliev [18] and Gschneidner and Verkade [19] were approximate ones, no thermodynamic parameters have been adjusted for (Ni, γCe) in the present work. In order to describe the temperature of the reaction: (δCe) ↔ (γCe) + liquid, one regular constant parameter was introduced for (δCe).

In the case of the stoichiometric intermetallic phases, it is possible to optimize independently the coefficients A and B in Eq. (6) since both enthalpy of formation and phase diagram data were available. In the literature, there are no experimental data about the enthalpies of formation for $CeNi_3$ and Ce_2Ni_7. Due to the complexity of the crystal structure, it is difficult to predict the enthalpy of formation using *ab initio* calculation method. However, the data for these two compounds ($CeNi_3$

Table 3
Summary of the thermodynamic parameters in the Ce–Ni system[a]

Liquid: (Ce, Ni)$_1$
$${}^{0}L^{\mathrm{liq}}_{\mathrm{Ce,Ni}} = -133692.0 + 9.2788T$$
$${}^{1}L^{\mathrm{liq}}_{\mathrm{Ce,Ni}} = 57986.3 - 10.5739T$$

(δCe): (Ce, Ni)$_1$
$${}^{0}L^{\mathrm{bcc}}_{\mathrm{Ce,Ni}} = -57465.9$$

Ce_7Ni_3: $Ce_{0.7}Ni_{0.3}$
$${}^{0}G^{\mathrm{Ce_7Ni_3}}_{\mathrm{Ce:Ni}} - 0.7\,{}^{0}G^{fcc}_{\mathrm{Ce}} - 0.3\,{}^{0}G^{fcc}_{\mathrm{Ni}} = -22279.0 - 0.1039T$$

CeNi: $Ce_{0.5}Ni_{0.5}$
$${}^{0}G^{\mathrm{CeNi}}_{\mathrm{Ce:Ni}} - 0.5\,{}^{0}G^{fcc}_{\mathrm{Ce}} - 0.5\,{}^{0}G^{fcc}_{\mathrm{Ni}} = -30810.7 - 1.8361T$$

$CeNi_2$: (Ce, Ni)$_{1/3}$(Ce, Ni)$_{2/3}$
$${}^{0}G^{\mathrm{CeNi_2}}_{\mathrm{Ce:Ce}} = 5000 + {}^{0}G^{fcc}_{\mathrm{Ce}}$$
$${}^{0}G^{\mathrm{CeNi_2}}_{\mathrm{Ce:Ni}} - (1/3)\,{}^{0}G^{fcc}_{\mathrm{Ce}} - (2/3)\,{}^{0}G^{fcc}_{\mathrm{Ni}} = -38032.2 + 4.3937T$$
$${}^{0}G^{\mathrm{CeNi_2}}_{\mathrm{Ni:Ce}} - (2/3)\,{}^{0}G^{fcc}_{\mathrm{Ce}} - (1/3)\,{}^{0}G^{fcc}_{\mathrm{Ni}} = 48032.2 - 4.3937T$$
$${}^{0}G^{\mathrm{CeNi_2}}_{\mathrm{Ni:Ni}} = 5000 + {}^{0}G^{fcc}_{\mathrm{Ni}}$$

$CeNi_3$: $Ce_{0.25}Ni_{0.75}$
$${}^{0}G^{\mathrm{CeNi_3}}_{\mathrm{Ce:Ni}} - 0.25\,{}^{0}G^{fcc}_{\mathrm{Ce}} - 0.75\,{}^{0}G^{fcc}_{\mathrm{Ni}} = -35426.4 + 3.1847T$$

Ce_2Ni_7: $Ce_{2/9}Ni_{7/9}$
$${}^{0}G^{\mathrm{Ce_2Ni_7}}_{\mathrm{Ce:Ni}} - (2/9)\,{}^{0}G^{fcc}_{\mathrm{Ce}} - (7/9)\,{}^{0}G^{fcc}_{\mathrm{Ni}} = -34713.9 + 3.3593T$$

$CeNi_5$: (Ce, Ni)$_{1/6}$(Ce, Ni)$_{5/6}$
$${}^{0}G^{\mathrm{CeNi_5}}_{\mathrm{Ce:Ce}} = 5000 + {}^{0}G^{fcc}_{\mathrm{Ce}}$$
$${}^{0}G^{\mathrm{CeNi_5}}_{\mathrm{Ce:Ni}} - (1/6)\,{}^{0}G^{fcc}_{\mathrm{Ce}} - (5/6)\,{}^{0}G^{fcc}_{\mathrm{Ni}} = -39586.7 + 128.62T$$
$$-23.85T\ln(T) - 0.00446333T^2 - 30916.7T^{-1}$$
$${}^{0}G^{\mathrm{CeNi_5}}_{\mathrm{Ni:Ce}} - (5/6)\,{}^{0}G^{fcc}_{\mathrm{Ce}} - (1/6)\,{}^{0}G^{fcc}_{\mathrm{Ni}} = 49586.7 - 128.62T$$
$$+23.85T\ln(T) + 0.00446333T^2 + 30916.7T^{-1}$$
$${}^{0}G^{\mathrm{CeNi_5}}_{\mathrm{Ni:Ni}} = 5000 + {}^{0}G^{fcc}_{\mathrm{Ni}}$$
$${}^{0}L^{\mathrm{CeNi_5}}_{\mathrm{Ce,Ni:Ce}} = {}^{0}L^{\mathrm{CeNi_5}}_{\mathrm{Ce,Ni:Ni}} = -1143.7$$
$${}^{0}L^{\mathrm{CeNi_5}}_{\mathrm{Ce:Ce,Ni}} = {}^{0}L^{\mathrm{CeNi_5}}_{\mathrm{Ni:Ce,Ni}} = -35496.0$$

[a] In J/(mole of atoms); temperatures (T) in Kelvin. The Gibbs energies for the pure elements are from the SGTE compilation [39].

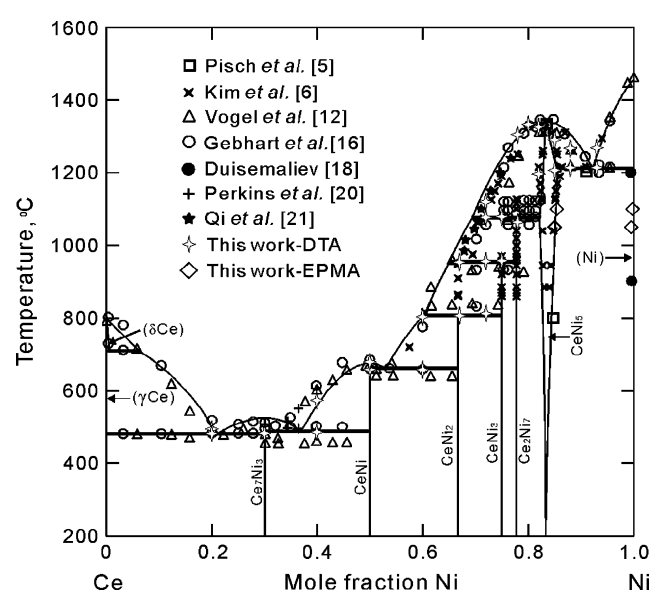

Fig. 1. Calculated Ce-Ni phase diagram, compared with the experimental data from the present work and the literature.

Table 4
Comparison between the measured and calculated invariant equilibria in the Ce–Ni system

Reaction	Composition, at.% Ni			Temperature (°C)	Source[a]
(δCe) ↔ (γCe) + L	0.58	0	6.2	710	This work*
L ↔ (γCe) + Ce_7Ni_3	21.0	0	30.0	484	[4]+
	19.1	0	30.0	470	[12]+
	22.0	0	30.0	477	[16,19]+
	22.5	0	30.0	475	[17]+
	–	0	30.0	481	This work+
	21.0	0	30.0	483	This work*
L ↔ Ce_7Ni_3 + CeNi	33.0	30.0	50.0	498	[4]+
	32.8	30.0	50.0	450	[12]+
	33.0	30.0	50.0	495	[16,19]+
	38.5	30.0	50.0	495	[17]+
	34.4	30.0	50.0	490	[20]+
	–	30.0	50.0	487	This work+
	36.9	30.0	50.0	487	This work*
L ↔ CeNi + $CeNi_2$	–	50.0	66.7	655	[4]+
	52.9	50.0	66.7	635	[12]+
	55.0	50.0	66.7	655	[16,19]+
	53.0	50.0	66.7	635	[17]+
	–	50.0	66.7	658	This work+
	53.1	50.0	66.7	658	This work*
L + $CeNi_3$ ↔ $CeNi_2$	–	75.0	66.7	830	[12,17,19]+
	62.0	75.0	66.7	830	[16]+
	–	75.0	66.7	813	This work+
	60.3	75.0	66.7	811	This work*
L + Ce_2Ni_7 ↔ $CeNi_3$	–	77.8	75.0	960	[12,17,19]+
	69.0	77.8	75.0	930	[16]+
	–	77.8	75.0	955	This work+
	65.1	77.8	75.0	957	This work*
L + $CeNi_5$ ↔ Ce_2Ni_7	73.7	82.4	77.8	1150	[6]+
	–	83.3	77.8	1065	[12,17]+
	72.0	83.3	77.8	1065	[16]+
	–	83.3	77.8	1055	[19]+
	–	82.6	77.8	1077	This work+
	71.0	82.6	77.8	1076	This work*
L ↔ $CeNi_5$ + (Ni)	91.1	–	–	1204	[5]+
	91.1	86.5	1.0	1215	[6]+
	91.1	83.3	1.0	1210	[12]+
	91.0	83.3	1.0	1205	[16]+
	91.0	83.3	1.0	1210	[17,19]+
	91.9	–	1.0	1215	This work+
	92.2	85.6	1.0	1211	This work*

[a] + Denotes the experimental results; * denotes the calculated results.

and Ce_2Ni_7) could be estimated from the line connecting those of $CeNi_2$ and $CeNi_5$.

For $CeNi_5$, two separate treatments are used in the present optimization. The first treatment is simply to consider $CeNi_5$ as the stoichiometric compound. In the second treatment, the two-sublattice model was introduced to describe the homogeneity range of $CeNi_5$ [6]. Yamaguchi et al. [7] obtained the following expression for $H_T - H_{298.15}$, $143.1T + 2.678 \times 10^4 T^2 + 3.7 \times 10^{-5} T^{-1} - 46300$ (J/mole of formula unit), for the alloy with the nominal composition of $CeNi_{5.13}$ using a drop calorimeter. The coefficients c–e in Eq. (8) are related to this expression by the following equations: $c = -143.1$, $d = -2.678 \times 10^{-2}$, $e = -1.855 \times 10^5$.

The thermodynamic parameters finally obtained in the present work are listed in Table 3. Compared with the parameters by Pisch et al. [5], all of the current parameters are introduced according to the types of experimental data and less number of adjustable coefficients is used in the present optimization. For example, in the case of liquid, only four parameters are used in the current modeling rather than six by Pisch et al. [5].

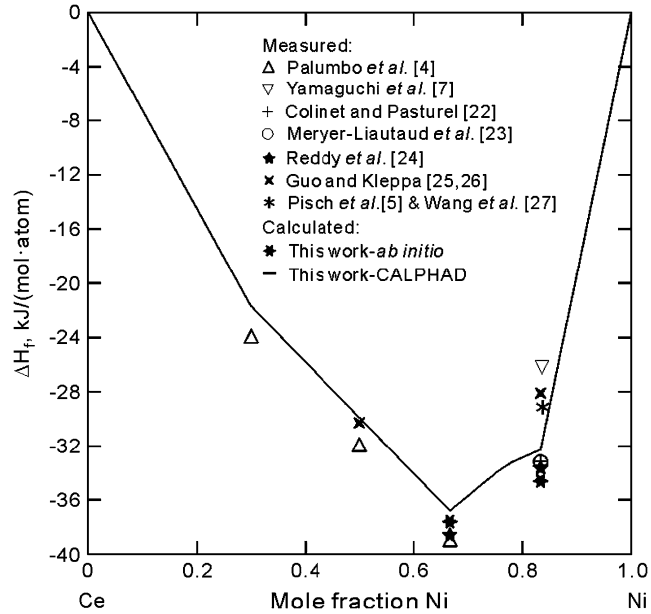

Fig. 2. Calculated and measured enthalpies of formation at 298.15 K.

A comparison of the computed Ce–Ni phase diagram with the experimental data is made in Fig. 1. As shown in this figure, the experimental data can be well reproduced by the modeling.

The comparisons between the measured and calculated invariant equilibria are listed in Table 4, showing also a good agreement.

Fig. 2 shows the enthalpies of formation resulting from measurement, *ab initio* calculation and thermodynamic modeling.

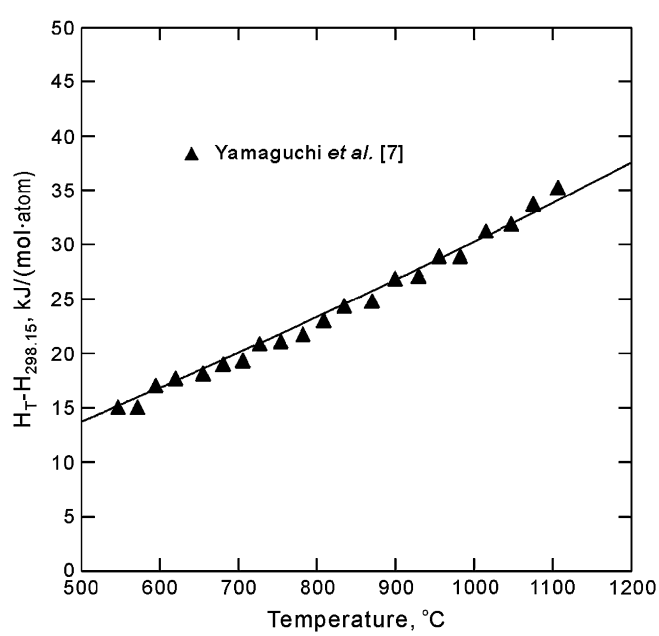

Fig. 3. Calculated heat content for $CeNi_{5.13}$, compared with the experimental data [7].

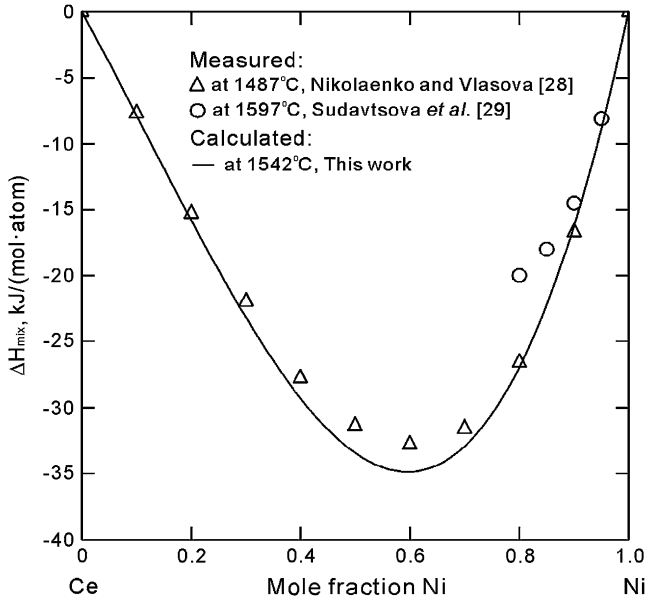

Fig. 4. Calculated integral enthalpy of mixing for the liquid phase at 1542 °C, compared with the experimental data of Nikolaenko and Vlasova [28] and Sudavtsova et al. [29].

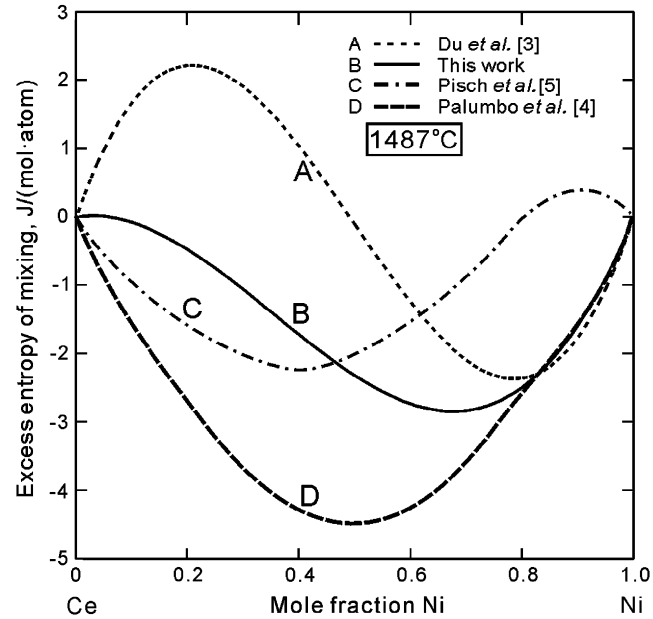

Fig. 6. Comparison of calculated excess entropy of mixing in the liquid phase at 1487 °C.

The *ab initio* calculated energies of formation of $CeNi_2$ and $CeNi_5$ are −37.6 and −34.6 kJ mol^{-1} atom^{-1}, respectively. Although the calculated data refer to 0 K, as a first approximation, it is reasonable to compare the 0 K data with the enthalpy of formation at room temperature.

A good consistency between the experimental and calculated enthalpies of formation could be observed for Ce_7Ni_3, CeNi, and $CeNi_3$.

With regard to the enthalpy of formation of $CeNi_5$, the calculated values from *ab initio* calculation and CALPHAD approach agree well with the ones obtained by Colinet and Pasturel [22], Meyer-Liautaud et al. [23] and Reddy et al. [24]. Although Wang et al. [27] claimed that the X-ray diffraction pattern of their nominal $CeNi_{5.2}$ alloy shows the appearance of a single $CeNi_5$ phase, the reported enthalpy of formation of the ideally stoichiometric $CeNi_5$ is considered to be approximate, because the discrepancy between nominal and practical compositions of the measured alloy has not been quantitatively determined. The less exothermic of the data from Yamaguchi et al. [7] and Guo and Kleppa [25] has been attributed to the contamination of the samples [24,25]. By means of XRD and SEM/EDX, Guo et al. [25] detected circa 2 at.% Ce_2O_3 accompanying the expected single $CeNi_5$ phase in the prepared alloys.

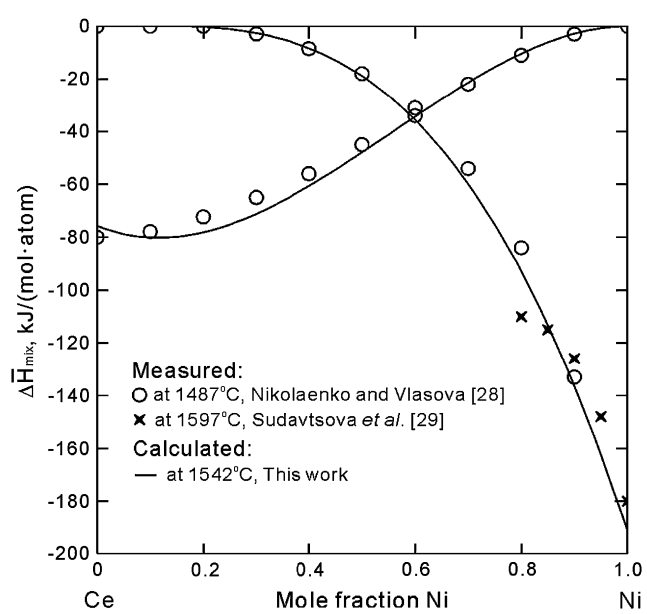

Fig. 5. Calculated partial enthalpy of mixing for the liquid phase at 1542 °C, compared with the experimental data of Nikolaenko and Vlasova [28] and Sudavtsova et al. [29].

In Fig. 3, the computed enthalpy increment for $CeNi_{5.13}$ is compared with the experimental data from Yamaguchi et al. [7]. The fit to the experimental data is fairly well.

Figs. 4 and 5 show the calculated integral and partial enthalpies of mixing in the liquid together with the experimental values [28,29], respectively. As shown in the figures, all of the experimental values are well reproduced by the present calculation, although they are not used in the evaluation of thermodynamic parameters. The optimized entropies at 298.15 K for $CeNi_2$ (40.77 J K^{-1} mol^{-1} atom^{-1}) and $CeNi_5$ (33.42 J K^{-1} mol^{-1} atom^{-1}) agree well with those of Wallace

Table 5
Comparison between calculated and measured enthalpies of reaction for the eutectic alloys in the Ce–Ni system

Reaction	ΔH (kJ mol^{-1} atom^{-1})		
	Measured	Calculated	
	[4]	[4]	This work
L ↔ (γCe) + Ce_7Ni_3	8.2 ± 0.5	8.1	7.5
L ↔ Ce_7Ni_3 + CeNi	8.4 ± 0.5	6.7	9.0
L ↔ CeNi + $CeNi_2$	9.0 ± 1.0	11.2	10.9

[30] (41.44 J K^{-1} mol^{-1} $atom^{-1}$) and Marzouk et al. [31] (35.64 J K^{-1} mol^{-1} $atom^{-1}$). These measured entropies are also not utilized in the present modeling. Thus, the good agreement between these model-predicted and measured values provides another check on the reliability of the present modeling.

The comparison of excess entropy of mixing of the liquid phase at 1487 °C from several calculations is shown in Fig. 6. The excess entropy of mixing of a metallic liquid is expected to be in the range of −10 to 5 J K^{-1} mol^{-1} $atom^{-1}$ [45]. This feature is observed for all the thermodynamic calculations [3–5].

Table 5 presents the comparison between the calculated and the measured enthalpies of fusion for three eutectic alloys. In general, the present modeling yields a better description of the measured enthalpies of fusion than the previous one [4]. Although Pisch et al. [5] mentioned this type of data, no comparison was made in their publication probably due to the irreproducibility of the quasi-chemical modeling.

The present description of the Ce–Ni system has been combined with the description of the Al–Ce [8] and Al–Ni [9] systems to calculate the Al–Ce–Ni ternary system successfully [46].

7. Conclusions

- Using 14 decisive alloys, which are analyzed with XRD, DTA, and EPMA techniques, the Ce–Ni phase diagram has been reinvestigated. *Ab initio* energetic calculations are employed to obtain the enthalpies of formation for $CeNi_2$ and $CeNi_5$ compounds.
- An optimal thermodynamic data set for the Ce–Ni system was obtained by considering the *ab initio* calculation values, present experimental results and critically assessed literature data. The comparisons show that the experimental information is accurately accounted for by the present description. Noticeable improvements have been made, compared with the pervious modeling.
- *Ab initio* energetic calculation for the thermodynamic property, such as enthalpy of formation, is an effective tool to verify the accuracy of measurements. A hybrid methodology of CALPHAD, *ab initio* calculation and key experiments in the thermodynamic investigation is highly recommended.

Acknowledgements

This research work is supported by National Natural Science Foundation of China (Grant No. 50571114) and National Outstanding Youth Science Foundation of China (Grant No. 50425103). The Thermo-Calc program developed by Thermo-Calc software Company of Sweden is gratefully acknowledged. Wei Xiong is grateful to Dr. Patrice E.A. Turchi (Lawrence Livermore National Laboratory, USA) for valuable discussions. One of the authors (Yong Du) acknowledges Cheung Kong Chair Professorship released by Ministry of Education of China and Furong Chair Professorship Program released by Hunan Province of P.R. China for financial support.

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